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References:

1. IoT Fundamentals: Networking Technologies, Protocols and Use Cases for Internet of Things, DavidHanes, Gonzalo Salgueiro, Patrick Grossetete, Rob Barton and Jerome Henry, Cisco Press, 2017
2. Internet of Things – A hands-on approach, Arshdeep Bahga, Vijay Madisetti, Universities Press, 2015

Faculty

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Dean Academics

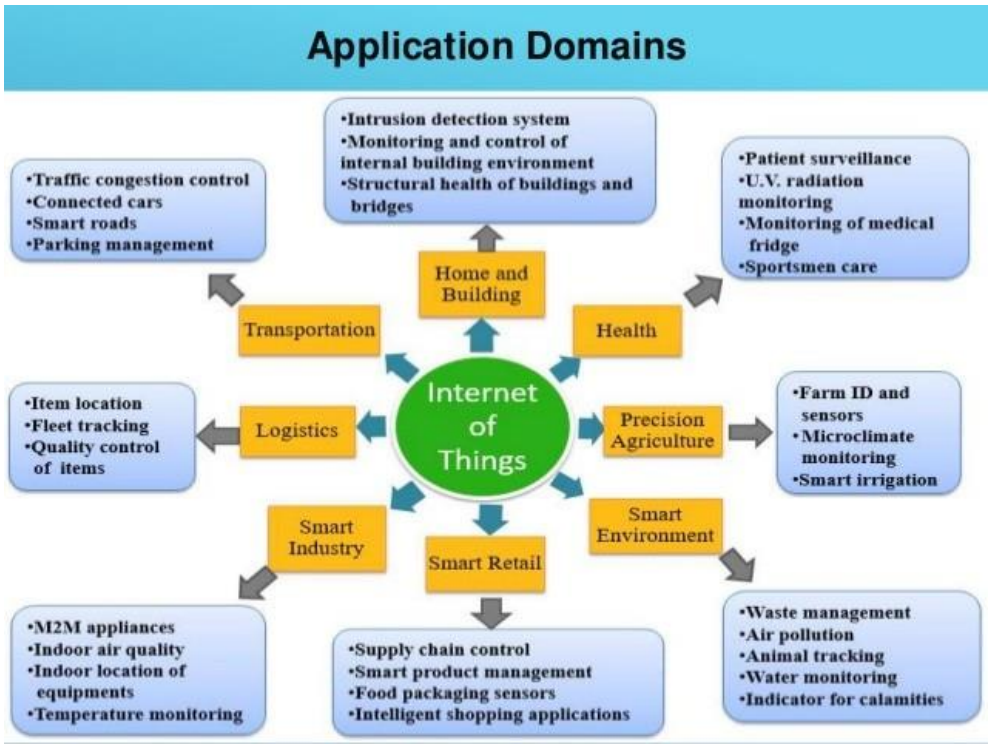
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IOT
Unit - 5
Lecture- 1

Domain specific IoTs:

Introduction:

The Internet of Things (IoT) applications span a wide range of domains including (but not limited to) homes, cities, environment, energy systems, retail, logistics, industry, agriculture and health. This chapter provides an overview of various typesof IoT applications for each of these domains. In the later chapters the reader is guided through detailed implementations of several of these applications.



Home Automation:

Smart Lighting:

Smart lighting for homes helps in saving energy by adapting the lighting to the ambient conditions and switching on/off or dimming the lights when needed. Key enabling technologies for smart lighting include solid state lighting (such as LED lights) and IP-enabled lights. For solid state lighting solutions both spectral and temporal characteristics can be configured to adapt illumination to various needs. Smart lighting solutions for home achieve energy savings by sensing the human movements and their environments and controlling the lights accordingly. Wireless-enabled and Internet connected lights can be controlled remotely from IoT applications such as a mobile or web application. Smart lights with sensors for occupancy, temperature, lux level, etc., can be configured to adapt the lighting (by changing the light intensity, color, etc.) based on the ambient conditions sensed, in order to provide a good ambiance. In controllable LED lighting system is presented that is embedded with ambient intelligence gathered from a distributed smart wireless sensor network to optimize and control the lighting system to be more efficient and user-oriented. A solid state lighting model is described in and implemented on a wireless sensor network that provides services for sensing illumination changes and dynamically adjusting luminary brightness according to user preferences. In chapter-9 we provide a case study on a smart lighting system,

Smart Appliances:

Modern homes have a number of appliances such as TVs, refrigerators, music systems, washer/dryers, etc. Managing and controlling these appliances can be cumbersome, with each appliance having its own controls or remote controls. Smart appliances make the management easier and also provide status information to the users remotely.

For example, smart washer/dryers that can be controlled remotely and notify whenthe washing/drying cycle is complete. Smart thermostats allow controlling the temperature remotely and can learn the user preferences. Smart refrigerators can keep track of the items stored (using RFID tags) and send updates to the users when an item is low on stock. Smart TVs allows users to search and stream videos and movies from the Internet on a local storage drive, search TV channel schedulesand fetch news, weather updates and other content from the Internet. OpenRemote is an open source automation platform for homes and buildings. OpenRemote is platform agnostic and works with standard hardware. With OpenRemote, users can control various appliances using mobile or web applications. OpenRemote comprises of three components - a Controller that manages scheduling and runtime integration between devices, a Designer that allows you to create both configurations for the controller and create user interface designs and Control Panels that allow you tointeract with devices and control them. An IoT-based appliance control system for smart homes that uses a smart central controller to set up a wireless sensor and actuator network and control modules for appliances.



Intrusion Detection:

Home intrusion detection systems use security cameras and sensors (such as PIR sensors and door sensors) to detect intrusions and raise alerts. Alerts can be in the form of an SMS or an email sent to the user. Advanced systems can even send detailed alerts such as an image grab or a short video clip sent as an email attachment. A cloud controlled intrusion detection system is described in that uses location-aware services, where the geo-location of each node of a home automation system is independently detected and stored in the cloud. In the event of intrusions, the cloud services alert the accurate neighbors (who are using the home automation system) or local police. An intrusion detection system based on UPnP technology is described. The system uses image processing to recognize the intrusion and extract the intrusion subject and generate Universal-Plug-and-Play (UPnP- based) instant messaging for alerts.

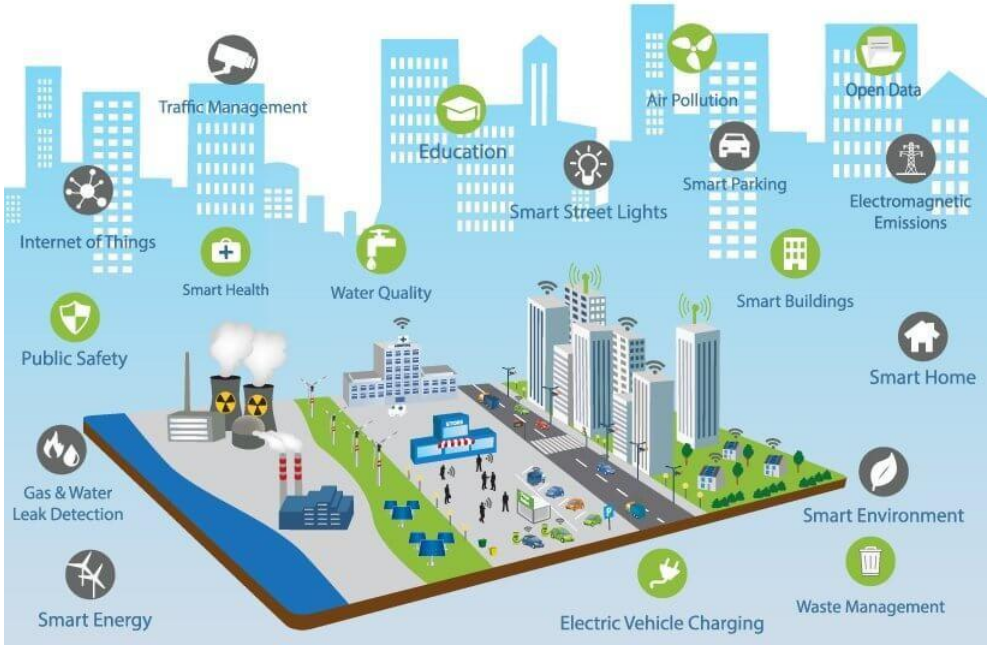
Smoke/Gas Detectors:

Smoke detectors are installed in homes and buildings to detect smoke that is typically an early sign of fire. Smoke detectors use optical detection, ionization or air sampling techniques to detect smoke. Alerts raised by smoke detectors can be in the form of signals to a fire alarm system, Gas detectors can detect the presence of harmful gases such as carbon monoxide (CO), liquid petroleum gas (LPG), etc. A smart smoke/gas detector can raise alerts in human voice describing where the problem is, send or an SMS or email to the user or the local fire safety department and provide visual feedback on its status (healthy, battery-low, etc.). The design of a system that detects gas leakage and smoke and gives visual level indication, is described.

Lecture 2

Cities:

The new Internet of Things (IoT) applications are enabling Smart City initiatives worldwide. It provides the ability to remotely monitor, manage and control devices, and to create new insights and actionable information from massive streams of real-time data. The main features of a smart city include a high degree of information technology integration and a comprehensive application of information resources. The essential components of urban development for a smart city should include smart technology, smart industry, smart services, smart management and smart life. The Internet of Things is about installing sensors (RFID, IR, GPS, laser scanners, etc.) for everything, and connecting them to the internet through specific protocols for information exchange and communications, in order to achieve intelligent recognition, location, tracking, monitoring and management. With the technical support from IoT, smart city need to have three features of being instrumented, interconnected and intelligent. Only then a Smart City can be formed by integrating all these intelligent features at its advanced stage of IOT development. The explosive growth of Smart City and Internet of Things applications creates many scientific and engineering challenges that call for ingenious research efforts from both academia and industry, especially for the development of efficient, scalable, and reliable Smart City based on IoT. New protocols, architectures, and services are in dire needs to respond for these challenges. The goal of the special issue is to bring together scholars, professors, researchers, engineers and administrators resorting to the state-of-the-art technologies and ideas to significantly improve the field of Smart City based on IoT.



Key Drivers for IoT Applications in Smart Cities:

There are several reasons for municipalities to move to the wireless communications methods offered by IoT technologies:

- **Cost** is a key driver in the decision to convert from wired solutions to wireless, as it is massively expensive to install and maintain landlines. Additionally, the costs of cellular data plans are dropping and the robustness and throughput of wireless communications are therefore enabling new use cases that would previously have been cost-prohibitive.

- **Efficiency** is another important impetus. In most wired solutions, service personnel must physically go to the installation site to audit and service the communications infrastructure. These ”truck rolls” are expensive and inefficient, since they typically occur on a schedule whether or not a problem exists. By contrast, wireless communications enable remote monitoring andmanagement of IoT deployments. This enables administrators to perform

firmware updates and security patches across the entire deployment, and get automated notifications in the event of any issues.

- **Resource reduction** is often a driver as well, particularly in use cases such as smart street lighting and monitoring assets. These IoT applications make it possible to use sensors to gather data and wireless modules to control resource use, which can result in a dramatic reduction in energy use.

Smart Lighting:

Smart lighting systems for roads, parks and buildings can help in saving energy. According to an IEA report. lighting is responsible for 19% of global electricity use and around 6% of global greenhouse gas emissions. Smart lighting allows lighting to be dynamically controlled and also adaptive to the ambient conditions. Smart lights connected to the Internet can be controlled remotely to configure lighting schedules and lighting intensity. Custom lighting configurations can be set for different situations such as a foggy day, a festival, etc. Smart lights equipped with sensors can communicate with other lights and exchange information on the sensed ambient conditions to adapt the lighting. The need for smart lighting system in smart cities, smart lighting features and how to develop interoperable smart lighting solutions.

Smart Roads:

Smart roads equipped with sensors can provide information on driving conditions, travel time estimates and alerts in case of poor driving conditions, traffic congestions and accidents, Such information can help in making the roads safer and help in reducing traffic jams. Information sensed from the roads can be communicated via Internet to cloud-based applications and social media and disseminated to the drivers who subscribe to such applications. A distributed and autonomous

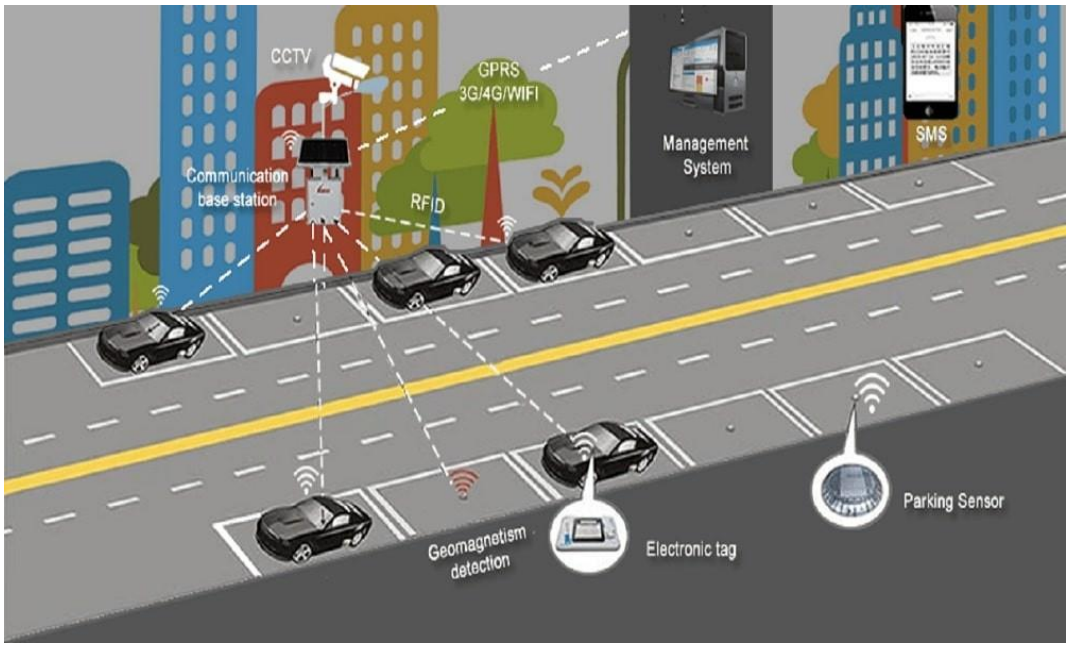
system of sensor network nodes for improving driving safety on public roads in proposed. The system can provide the drivers and passengers with a consistent view of the road situation a few hundred meters ahead of them or a few dozen miles away, so that they can react to potential dangers early enough.

Smart Parking:

Finding a parking space during rush hours in crowded cities can be time consumingand frustrating. Furthermore, drivers blindly searching for parking spaces create additional traffic tion. Smart parking make the search for parking space easier and convenient for Smart parking are powered by IoT systems that detect the number of empty parking slots and send the information over the Internet to smart parking application back-ends, These applications can be accessed by the drivers from smart-phones, tablets and in-car navigation systems. In smart parking, sensors are used for each parking slot, to detect whether the slot is empty or occupied. This information is aggregated by a local controller and then sent over the Internet to the database. Recent system describe latest trends in parking availability monitoring, parking reservation and dynamic pricing schemes. Design and implementation of a prototype smart parking system based on wireless sensor network technology with features like remote parking monitoring, automated guidance, and parking reservation.

Structural Health Monitoring:

Structural Health Monitoring i s a network of sensors to monitor the vibration levels in the structures such as bridges and buildings. The data collected from these sensors is analyzed to assess the health of the structures. By analyzing thedata it is possible to detect cracks and mechanical breakdowns, locate the damages



to a structure and also calculate the remaining life of the structure. Using such systems, advance warnings can be given in the case of imminent failure of the structure. Since structural health monitoring systems use large number of wireless sensor nodes which are powered by traditional batteries, researchers are exploring energy harvesting technologies to harvesting.

Surveillance:

Surveillance of infrastructure, public transport and events in cities is required to ensure safety and security. City wide surveillance infrastructure comprising of large number of distributed and Internet connected video surveillance cameras can be created. The video feeds from surveillance cameras can be aggregated in cloud- based scalable storage solutions.

Cloud-based video analytics applications can be developed to search for patterns

or specific events from the video feeds. A smart city surveillance system is described that leverages benefits of cloud data stores.

Emergency Response:

ToT systems can be used for monitoring the critical infrastructure in cities such as buildings, gas and water pipelines, public transport and power substations. IoT systems for fire detection, gas and water leakage detection can help in generating alerts and minimizing their effects on the critical infrastructure. IoT systems for critical infrastructure monitoring enable aggregation and sharing of information collected from large number of sensors. Using cloud-based architectures, multi-modal information such as sensor data, audio, video feeds can be analyzed in near real-time to detect adverse events. Response to alerts generated by such systems can be in the form of alerts sent to the public, re-routing of traffic, evacuationsof the affected areas, etc. The system adapts by dynamically adjusting traffic lights, changing related driving policies, recommending behavior change to drivers, and applying essential security controls. Such systems can reduce the latency of emergency services for vehicles such as ambulances and police cars while minimizing disruption of regular traffic.

Lecture 3

Environment:

Weather Monitoring:

IoT-based weather monitoring systems can collect data from a number of sensor attached (such as temperature, humidity, pressure, etc.) and send the data to cloud-based applications and storage back-ends. The data collected in the cloud can thenbe analyzed and visualized by cloud-based applications. Weather alerts can be sentto the subscribed users from such applications. AirPi is a weather and air qualitymonitoring kit capable of recording and uploading information about temperature,humidity, air pressure, light levels, UV levels, carbon monoxide, nitrogen dioxide and smoke level to the Internet. A pervasive weather monitoring system is described that is integrated with buses to measure weather variables like humidity,temperature and air quality during the bus path. A weather monitoring system based on wireless sensor networks is described.

Air Pollution Monitoring:

IoT based air pollution monitoring systems can monitor emission of harmful gases (CO, CO, NO, NOp, etc.) by factories and automobiles using gaseous and meteorologicalsensors. The collected data can be analyzed to make informed decisions on pollutions control approaches. A real-time air quality monitoring system is presented that comprises of several distributed monitoring stations that communicate via wireless with a back-end server using machine-to-machine communication. An air pollution system is described that integrates a single-chip microcontroller,several air pollution sensors, GPRS-Modem, and a GPS module.



Noise Pollution Monitoring:

Due to growing urban development, noise levels in cities have increased and even become alarmingly high in some cities, Noise pollution can cause health hazards for humans due to sleep disruption and stress. Noise pollution monitoring can help in generating noise maps for cities, Urban noise maps can help the policy makers inurban planning and making policies to control noise levels near residential areas,schools and parks. IoT based noise pollution monitoring systems use a number ofnoise monitoring stations that are deployed at different places in a city. The data on noise levels from the stations is collected on servers or in the cloud. The collecteddata is then aggregated to generate noise maps. In [43], a noise mapping study fora city is presented which revealed that the city suffered from serious noise pollution. In [44], the design of smart phone application is described that allows the users to continuously measure noise levels and send to a central server where all generated

information is aggregated and mapped to a meaningful noise visualization map.

Forest Fire Detection:

Forest fires can cause damage to natural resources, property and human life. There can be different causes of forest fires including lightening, human negligence, volcanic eruptions and sparks from rock falls. Early detection of forest fires can help in minimizing the damage. IoT based forest fire detection systems use a number of monitoring nodes deployed at different locations in a forest. Each monitoring node collects measurements on ambient conditions including temperature, humidity, light levels, etc. A system for early detection of forest fires that provides early warning of a potential forest fire and estimates the scale and intensityof the fire if it materializes. A forest fire detection system based on wireless sensor networks in presented. The system uses multi-criteria detection which is implemented by the artificial neural network (ANN). The ANN fuses sensing data corresponding to multiple attributes of a forest fire (such as temperature, humidity, infrared and visible light) to detect forest fires.

River Floods Detection:

River floods can cause extensive damage to the natural and human resources and human life. River floods occur due to continuous rainfall which cause the river levels to rise and flow rates to increase rapidly. Early warnings of floods can be given by monitoring the water level and flow rate. IoT based river flood monitoring system use a number of sensor nodes that monitor the water level (using ultrasonic sensors) and flow rate (using the flow velocity sensors). Data from a number of such sensor nodes is aggregated in a server or in the cloud. Monitoring applications raise alerts when rapid increase in water level and flow rate is detected. A river flood monitoring system in described that measures river and weather conditions through

wireless sensor nodes equipped with different sensors. A motes-based sensor network for river flood monitoring is described. The system includes a water level monitoring module, network video recorder module, and data processing module that provides flood information in the form of raw data, predicted data, and videofeed.

Lecture 4

Energy:

Smart Grids:

Smart Grid is a data communications network integrated with the electrical grid that collects and analyzes data captured in near-real-time about power transmission, distribution, and consumption. Smart Grid technology provides predictive information and recommendations to utilities, their suppliers, and their customers on how best to manage power. Smart Grids collect data regarding electricity generation (centralized or distributed), consumption (instantaneous or predictive), storage (or conversion of energy into other forms), distribution and equipment health data. Smart grids use high-speed, fully integrated, two-way communication technologies for real-time information and power exchange. By using loT based sensing and measurement technologies, the health of equipment and the integrity of the grid can be evaluated. Smart meters can capture almost real-time consumption, remotely control the consumptionof electricity and remotely switch off supply when required. Power thefts can be prevented using smart metering. By analyzing the data on power generation, transmission and consumption smart girds can improve efficiency throughout the electric system. Storage collection and analysis of smarts grids data in the cloud can help in dynamic optimization of system operations, maintenance, and planning. Cloud-based monitoring of smart grids data can improve energy usage levels via energy feedback to users coupled with real-time pricing information. Real-time demand response and management strategies can be used for lowering peak demand and overall load via appliance control and energy storage mechanisms. Conditionmonitoring data collected from power generation and transmission systems can help in detecting faults and predicting outages.

Smart Meter:

Smart metering benefits utilities by improving customer satisfaction with faster interaction, while giving consumers more control of their energy usage to save money and reduce carbon emissions. With power visibility all the way to the meter, utilities can optimize energy distribution and even take action to shift demand loads.

Smart metering helps utilities to:

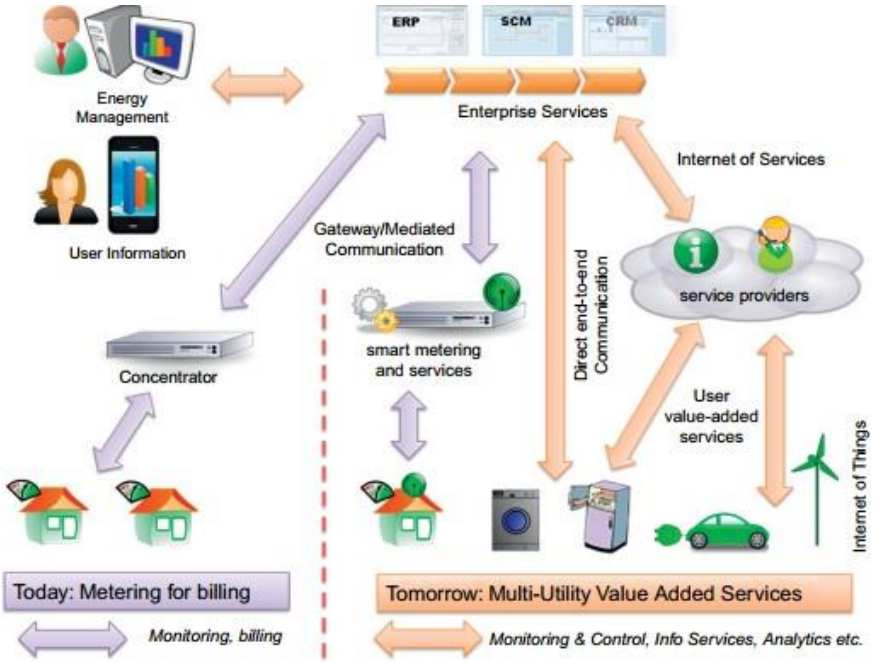
- Reduce operating expenses by managing manual operations remotely
- Improve forecasting and streamline power-consumption
- Improve customer service through profiling and segmentation
- Reduce energy theft
- Simplify micro-generation monitoring and track renewable power

In addition Smart Meters can enable an array of new smart services to residential, commercial, and industrial customers to better manage their energy usage patterns, reduce overall power consumption and even benefit from new pricing models.

Cellular communications provides a reliable connectivity option for smart metering infrastructure, including full IP infrastructure and low latency in 4G LTE. With the ubiquitous reach of modern cellular networks, and the development of LTE-M (LTE for M2M) providing long range low power cost effective solutions, utilities can connect meters easily and inexpensively virtually anywhere. And they can benefit from a proven, highly reliable communications infrastructure without taking on the costs of deploying and maintaining it themselves.

Renewable Energy Systems:

Due to the variability in the output from renewable energy sources (such as solar and wind), integrating them into the grid can cause grid stability and reliability



problems. Variable output produces local voltage swings that can impact power quality. Existing grids were designed to handle power flows from centralized generationsources to the loads through transmission and distribution lines. When distributed renewable energy sources are integrated into the grid, they create power bi-directional power flows for which the grids were not originally designed. IoT based systems integrated with the transformers at the point of interconnection measure the electrical variables and how much power is fed into the grid. To ensure the grid stability, one solution is to simply cut off the overproduction. For wind energy systems, closed- loop controls can be used to regulate the voltage at point of interconnection which coordinate wind turbine outputs and provides reactive power support .

Prognostics:

Energy systems (smart grids, power plants, wind turbine farms, for instance) have a large number of critical components that must function correctly so that the systems

can perform their operations correctly. For example, a wind turbine has a number of critical components, e.g., bearings, turning gears, for instance, that must be monitored carefully as wear and tear in such critical components or sudden changein operating conditions of the machines can result in failures. In systems such as power grids, real-time information is collected using specialized electrical sensors called Phasor Measurement Units (PMU) at the substations. The informationreceived from PMUs must be monitored in real-time for estimating the state of thesystem and for predicting failures. Energy systems have thousands of sensors that gather real-time maintenance data continuously for condition monitoring and failure prediction purposes. IoT based prognostic real-time health management systems can predict performance of machines or energy systems by analyzing the extent of deviation of a system from its normal operating profiles. Analyzing massive amountsof maintenance data collected from sensors in energy systems and equipment can provide predictions for the impending ures (potentially in real-time) so that their reliability and availability can be improved. Prognostic health management systems have been developed for different energy systems. OpenPDC is a set of applicationsfor processing of streaming time-series data collected from Phasor Measurement Units (PMUs) in real-time. A generic framework for storage, processing and analysis of massive machine maintenance data, collected from a large number of sensors embedded in industrial machines, in a cloud computing environment was proposed in.

Smart light, temperature, air condition control:

The most obvious way to save energy is to cut down on wasting it. Smart lighting, learning thermostats and sensor-based HVAC systems of the new generation are designed to automatically maintain the perfect conditions in spaces and keep energy use at the optimum level. Equipped with different sensors (light, movement, humidity,

CO2 level, etc.) these systems can dynamically adjust the regimes depending on the changing conditions and avoid using energy in potential waste points. Thus, smart lighting dims depending on the amount of daylight available in the room and automatically turns off once the room is empty. Learning thermostats precool the space before the heat comes to avoid using expensive energy at the peak load time.

Energy Storage System:

In the case of smart storage, for example, the residents can make informed decisions no the loads and choose which loads should be protected, how much energy should bespent in the off-grid mode and where. Moreover, using smart storage, the adopters of solar and other renewables can now better manage the clean energy they generate, control the surplus and ensure maximized performance for their power network. In other words, energy storage is the application of Internet of Things in energy management which enables saving energy both directly and indirectly.

Connected stations, plants, solar and wind fields:

Another application of IoT for saving energy is focused on optimizing the very source of power production. Stations, plants, solar fields and wind turbines also consume energy, require maintenance and a wide range of effort and resource-heavy worksto keep them running. Using IoT in this sector, or Internet of Energy Things, is the right way to maximize the performance. Resource management in Internet of Energy Things implies a complex of measures to optimize the performance at a power grid. It includes using sensors, data analytics, predictive maintenance and other practices. Continuous condition monitoring of equipment and wiring using sensors, for example, prevents overloads and helps maintain a balanced load on the line. Predictive maintenance ensures timely repair of the equipment and prevents blackouts, accidents and costly downtime. In some cases, connecting power plants

and renewable grids into a network gives consumers a transparent view of where their energy comes from. And with this information, consumers get a choice to usethe cleanest source available at the moment.

Lecture 5

Retail and Logistics:

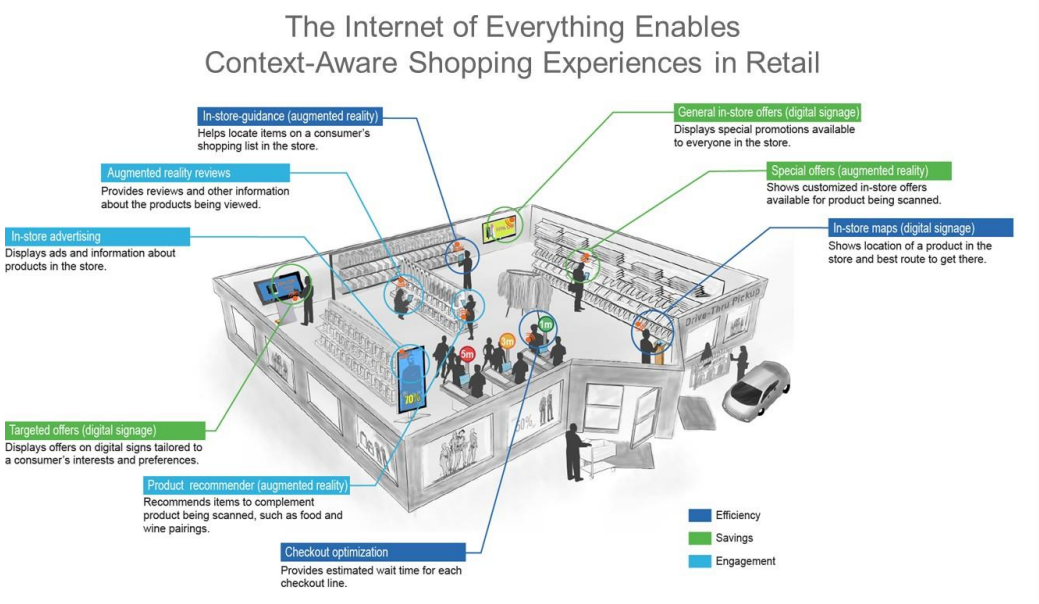
Retail:

Inventory Management:

Inventory management for retail has become increasingly important in the recent years with the growing competition. While over-stocking of products can result in additional storage expenses and risk (in case of perishables), under-stocking can lead to loss of revenue. loT systems using Radio Frequency Identification (RFID)tags can help in inventory management and maintaining the right inventory levels. RFID tags attached to the products allow them to be tracked in real-time so thatthe inventory levels can be determined accurately and products which are low on stock can be replenished. Tracking can be done using RFID readers attached to the retail store shelves or in the warehouse. IoT systems enable remote monitoring of inventory using the data collected by the RFID readers. An RFID data-based inventory management system for time-sensitive materials is described.

Smart Payments:

Smart payment solutions such as contact-less payments powered by technologies such as Near field communication (NFC) and Bluetooh, Near field communication (NFC) is a set of standards for smart-phones and other devices to communicate with eachother by bringing them into proximity or by touching them. Customers can store the credit card information in their NFC-enabled smart-phones and make paymentsby bringing the smart-phones near the point of sale terminals. NFC maybe used in combination with Bluetooh, where NFC (which offers low speeds) initiates initial pairing of devices to establish a Bluetooh connection while the actual data transfer



takes place over Bluetooh. The applications of NFC for contact-less payments are described in: .

Smart Vending Machines:

Smart vending machines connected to the Internet allow remote monitoring of inventory levels, elastic pricing of products, promotions, and contact-less paymentsusing NFC. Smart-phone applications that communicate with smart vending machinesallow user preferences to be remembered and learned with time. When a user moves from one vending machine to the other and pairs the smart-phone with the vending machine, a user specific interface is presented. Users can save their preferences and favorite products. Sensors in a smart vending machine monitor its operations and send the data to the cloud which can be used for predictive maintenance. Smart vending machines can communicate with other vending machines in their vicinity and share their inventory levels so that the customers can be routed to the nearest machine in case a product goes out of stock in a machine. For perishable items, the smart vending machines can reduce the price as the expiry date nears.

New products can be recommended to the customers based on the purchase history and preferences.

Logistics:

Route Generation & Scheduling:

Modern transportation systems are driven by data collected from multiple sources which is processed to provide new services to the stakeholders, By collecting large amount of data from various sources and processing the data into useful information, data-driven transportation systems can provide new services such as advanced route guidance, dynamic vehicle routing , anticipating customer demands for pickup and delivery problem, for instance, Route generation and scheduling systems can generate end-to-end routes using combination of route patterns and transportation modes and feasible schedules based on the availability of vehicles. As the transportation network grows in size and complexity, the number of possible route combinations increases exponentially. IoT based systems backed by the cloudcan provide fast response to the route generation queries and can be scaled up to serve a large transportation network.

Fleet Tracking:

Vehicle fleet tracking systems use GPS technology to track the locations of the vehicle real-time. Cloud-based fleet tracking systems can be scaled up on demand to handle large number of vehicles. Alerts can be generated in case of deviationsin planned routes. The vehicle locations and routes data can be aggregated and analyzed for detecting bottlenecks in the supply chain such as traffic congestions on routes, assignments and generation of alternative routes, and supply chain optimization. A fleet tracking system for commercial vehicles is described. The system

can analyze messages sent from the vehicles to identify unexpected incidents and discrepancies between actual and planned data, so that remedial actions can be taken.



Shipment Monitoring:

Shipment monitoring solutions for transportation systems allow monitoring the conditions inside containers. For example, containers carrying fresh food produce can be monitored to prevent spoilage of food. IoT based shipment monitoring systems use sensors such as temperature, pressure, humidity, for instance, to monitor the conditions inside the containers and send the data to the cloud, where it canbe analyzed to detect food spoilage. The analysis and interpretation of data onthe environmental conditions in the container and food truck positioning can enable

more effective routing decisions in real time. Therefore, it is possible to take remedial measures such as - the food that has a limited time budget before it gets rotten canbe re-routed to a closer destinations, alerts can be raised to the driver and the distributor about the transit conditions, such as container temperature exceeding the allowed limit, humidity levels going out of the allowed limit, for instance, and corrective actions can be taken before the food gets damaged. A cloud-based framework for real-time fresh food supply tracking and monitoring was proposedin . For fragile products, vibration levels during shipments can be tracked using accelerometer and gyroscope sensors attached to loT devices. The system monitors the vibration patterns of a container and its contents to reveal information relatedto its operating environment and integrity during transport, handling and storage.

Remote Vehicle Diagnostics:

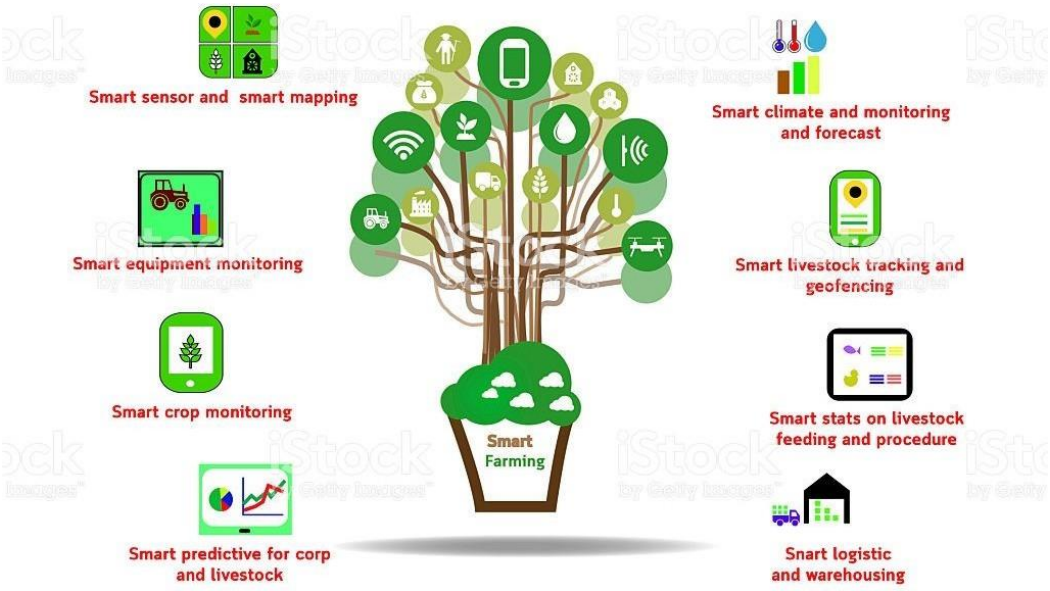
Remote vehicle diagnostic systems can detect faults in the vehicles or warn of impending faults. These diagnostic systems use on-board IoT devices for collecting data on vehicle operation (such as speed, engine RPM, coolent temperature, fault code number) and status of various vehicle sub-systems. Such data can be capturedby integrating on-board diagnostic systems with IoT devices using protocols such as CAN bus. Modern commercial vehicles support on-board diagnostic (OBD) standards such as OBD-II. OBD systems provide real-time data on the status of vehicle stems and diagnostic trouble codes which allow rapidly identifying the faultsin the vehicle. IoT based vehicle diagnostic systems can send the vehicle data tocentralized servers or the cloud where it can be analyzed to generate alerts andsuggest remedial actions. The system makes use of on-board vehicle diagnostics device and expert system to achieve real-time vehicle diagnostics and fault warning.

Lecture 6

Agriculture:

Smart Irrigation:

Smart irrigation systems can improve crop yields while saving water. Smart irrigation systems use IoT devices with soil moisture sensors to determine the amount of moisture in the soil and release the flow of water through the irrigation pipes only when the moisture levels go below a predefined threshold. Smart irrigation systems also collect moisture level measurements on a server or in the cloud where the collected data can be analyzed to plan watering schedules.



Green House Control:

Green houses are structures with glass or plastic roofs that provide conducive environment for growth of plants. The climatological conditions inside a green house can be monitored and controlled to provide the best conditions for growth of plants. The temperature, humidity, soil moisture, light and carbon dioxide levels are monitored using sensors and the climatological conditions are controlled automatically using actuation devices (such as valves for releasing water and switches for controlling fans). IoT systems play an important role in green house control and help in improving productivity. The data collected from various sensors is stored on centralized servers or in the cloud where analysis is performed to optimize the control strategies and also correlate the productivity with different control strategies.The design of a wireless sensing and control system for precision green house managementis described. The system uses wireless sensor network to monitor and control the agricultural parameters like temperature and humidity in real time for better management and maintenance of agricultural production.

Climate Conditions:

Climate plays a very critical role for farming. And having improper knowledge about climate heavily deteriorates the quantity and quality of the crop production.But IoT solutions enable you to know the real-time weather conditions. Sensors are placed inside and outside of the agriculture fields. They collect data from the environment which is used to choose the right crops which can grow and sustain inthe particular climatic conditions. The whole IoT ecosystem is made up of sensorsthat can detect real-time weather conditions like humidity, rainfall, temperature and more very accurately. There are numerous no. of sensors available to detect all these parameters and configure accordingly to suit your smart farming requirements. These sensors monitor the condition of the crops and the weather surrounding

them. If any disturbing weather conditions are found, then an alert is send. What gets eliminated is the need of the physical presence during disturbing climatic conditions which eventually increases the productivity and help farmers to reap more agriculture benefits.

Precision Farming:

Precision Agriculture/Precision Farming is one of the most famous applications of IoT in Agriculture. It makes the farming practice more precise and controlled by realizing smart farming applications such as livestock monitoring, vehicle tracking, field observation, and inventory monitoring. The goal of precision farming is to analyze the data, generated via sensors, to react accordingly. Precision Farming helps farmers to generate data with the help of sensors and analyze that informationto take intelligent and quick decisions. There are numerous precision farming techniques like irrigation management, livestock management, vehicle tracking and many more which play a vital role in increasing the efficiency and effectiveness. With the help of Precision farming, you can analyze soil conditions and otherrelated parameters to increase the operational efficiency. Not only this you can alsodetect the real-time working conditions of the connected devices to detect water and nutrient level.

Crop management:

One more type of IoT product in agriculture and another element of precision farming are crop management devices. Just like weather stations, they should be placed in the field to collect data specific to crop farming; from temperature and precipitation to leaf water potential and overall crop health.Thus, you can monitor your crop growth and any anomalies to effectively prevent any diseases or infestations that can harm your yield.

Cattle monitoring and management:

Just like crop monitoring, there are IoT agriculture sensors that can be attachedto the animals on a farm to monitor their health and log performance. Livestock tracking and monitoring help collect data on stock health, well-being, and physical location. For example, such sensors can identify sick animals so that farmers can separate them from the herd and avoid contamination. Using drones for real-time cattle tracking also helps farmers reduce staffing expenses.

Predictive analytics for smart farming:

Precision agriculture and predictive data analytics go hand in hand. While IoT and smart sensor technology are a goldmine for highly relevant real-time data, theuse of data analytics helps farmers make sense of it and come up with important predictions: crop harvesting time, the risks of diseases and infestations, yield volume, etc. Data analytics tools help make farming, which is inherently highly dependenton weather conditions, more manageable, and predictable.

For example, the Crop Performance platform helps farmers access the volume and quality of yields in advance, as well as their vulnerability to unfavorable weather conditions, such as floods and drought. It also enables farmers to optimize the supply of water and nutrients for each crop and even select yield traits to improve quality.

DRONES Monitoring:

Agriculture is one of the major industries to incorporate drones. Drones equipped with sensors and cameras are used for imaging, mapping and surveying the farms. There are ground based drones and aerial drones. Ground drones are bots that survey the fields on wheels. Aerial drones- formally known as unmanned aerial vehicles (UAVs) or unmanned aircraft systems (UASes) are flying robots. Drones can be remotely controlled remotely or they can fly automatically through software-

controlled flight plans in their embedded systems, working in coordination with sensors and GPS. From the drone data, insights can be drawn regarding crop health, irrigation, spraying, planting, soil and field, plant counting and yield prediction and much more. Drones can either be scheduled for farm surveys (drone as a service) or can be bought and stored near farms where they can be recharged and maintained. After the surveys the drones need to be taken to nearby labs to analyse the datathat has been collected.

Remote Sensing:

IoT based remote sensing utilizes sensors placed along the farms like weather stations for gathering data which is transmitted to analytical tool for analysis. Sensorsare devices sensitive to anomalies. Farmers can monitor the crops from analytical dashboard and take action based on insights.

- **Crop Monitoring:** Sensors placed along the farms monitor the crops for changes in light , humidity, temperature, shape and size. Any anomaly is detected by the sensors is analysed and farmer is notified. Thus remote sensing can help prevent the spread of diseases and keep an eye on the growth of crops.

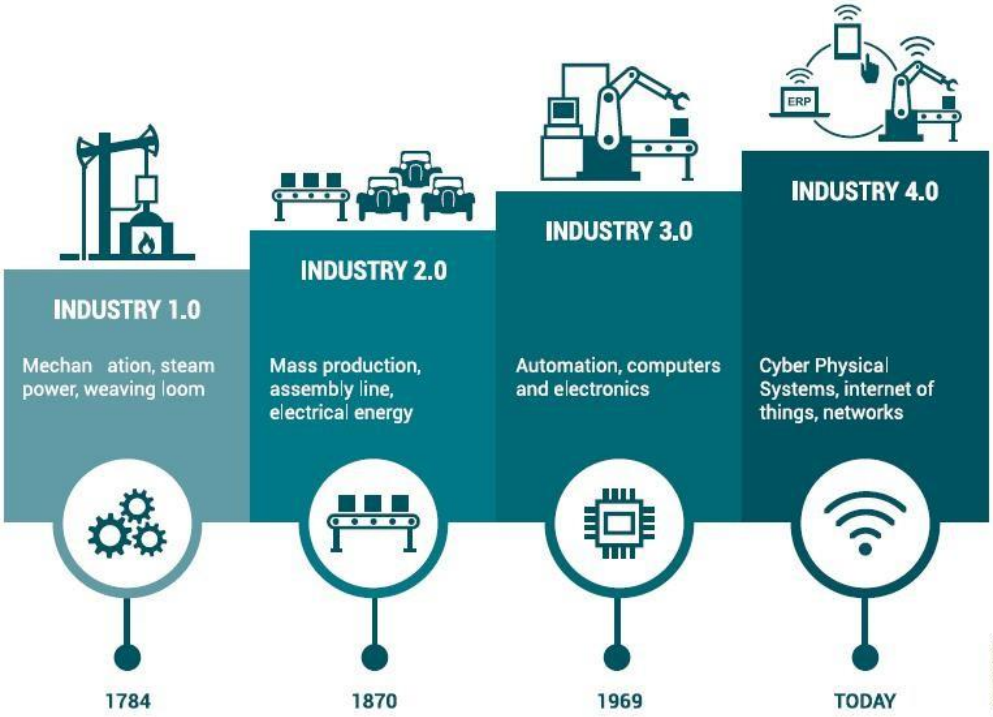
- **Weather conditions:** The data collected by sensors in terms of humidity, temperature, moisture precipitation and dew detection helps in determining the weather pattern in farms so that cultivation is done for suitable crops.

- **Soil quality:** The analysis of quality of soil helps in determining the nutrient value and drier areas of farms, soil drainage capacity or acidity, which allows to adjust the amount of water needed for irrigation and the opt most beneficial type of cultivation.

Lecture 7

Industry:

The Industrial Internet of Things (IIoT) takes networked sensors and intelligent devices and puts those technologies to use directly on the manufacturing floor, collecting data to drive artificial intelligence and predictive analytics.



The IoT is driving unprecedented disruption in an industry that has struggled in recent years due to talent shortages, and this offers hope for the industry?s future. The IIoT can transform traditional, linear manufacturing supply chains into dynamic, interconnected systems?a digital supply network (DSN)?that can more readily incorporate ecosystem partners. As key enablers of DSNs, IIoT technologieshelp to change the way that products are made and delivered, making factories more efficient, ensuring better safety for human operators, and, in some cases, saving

millions of dollars.

Machine Diagnosis & Progno:

Machine prognosis refers to predicting the performance of a machine by analyzing the data on the current operating conditions and how much deviations exist fromthe normal operating conditions. Machine diagnosis refers to determining the causeof a machine fault. IoT plays a major role in both prognosis and diagnosis of industrial machines. Industrial machines have a large number of components thatmust function correctly for the machine to perform its operations. Sensors in machines can monitor the operating conditions such as (temperature and vibration levels). The sensor data measurements are done on timescales of few millisecondsto few seconds, which leads to generation of massive amount of data. IoT based systems integrated with cloud-based storage and analytics back-ends can help in storage, collection and analysis of such massive scale machine sensor data. A number of methods

have been proposed for reliability analysis and fault prediction in machines. Case-based reasoning (CBR) is a commonly used method that finds solutions to new problems based on past experience, This past experience is organized and represented as cases in a case-base. CBR is an effective technique for problem solving in the fields in which it is hard to a quantitative mathematical model, such as machine diagnosis and prognosis. Since for each machine, data from a very largenumber of sensors is collected, using such high dimensional data for creation of case library reduces the case retrieval efficiency. Therefore, data reduction and feature extraction methods are used to find the representative set of features which have the same classification ability as the complete of features.

Power management: Building automation lowers energy costs:

Manufacturing, by its very nature, requires a lot of energy, and that in turn can account for a large percentage of operating costs. That?s why factory owners and managers are increasingly turning to IoT-based building management systems to connect sensors, actuators, controllers and other equipment over one IP backbone, enabling the monitoring of energy usage, lighting, HVAC and fire safety systems. This data can also be combined with information from broader datasets such as weather forecasting and financial information - such as the price of electricity and other utilities - to give a far more rounded view of building management. This type of architecture is finding increasing adoption in manufacturing environments, to make buildings smarter, more sustainable and more efficient.

Power management: Building automation lowers energy costs:

Faster and more flexible production lines might be the key to meeting customer demand, but there can also be a negative impact on quality control if monitoringisn?t up to scratch. These days, as plants look to automation to replace tasks suchas manual inspection, new technology is being used to ensure there is no deviation from quality parameters. Increasingly, the function of replacing the human eye has been performed by IoT-enabled high-pixel camera vision systems in combination with other devices such as acoustic sensors, along with high-performance image processing software. These can be used to identify defects such as size, shape or finish, and to check the accuracy and readability of labels, barcodes or QR codes.This information can then be looped back to earlier stages in the production lineallowing production managers to identify and classify the root cause of the problem before rectification action can be taken. Over time, artificial intelligence can be applied to learn from feedback and continuously refine and improve the production process.

This type of vision system is being used across manufacturing to monitor the quality of a wide range of products including electronic devices, consumer goods and metal finished parts.

Indoor Air Quality Monitoring:

Monitoring indoor air quality in factories is important for health and safety of the workers. Harmful and toxic gases such as carbon monoxide (CO), nitrogen monoxide (NO), Nitrogen Dioxide (NO), etc., can cause serious health problems. IoT based gas monitoring systems can help in monitoring the indoor air quality using variousgas sensors. The indoor air quality can vary for different locations. Wireles: sor networks based IoT devices can identify the hazardous zones, so that corrective measures can be taken to ensure proper ventilation. A hybrid sensor system for indoor air quality monitoring is presented, which contains both stationary sensors (for accurate readings and calibration) and mobile sensors (for coverage). A wire solution for indoor air quality monitoring is described that measures the environmental parameters like temperature, humidity, gaseous pollutants, aerosol and particulate matter to determine the indoor air quality.

Indoor Air Quality Monitoring:

A small to medium-sized manufacturing plant might contain hundreds of operator tools, in various shapes and sizes, which are used for a multitude of functions. For a large factory, that number could rise to thousands. Now imagine if all of those hammers, drills, torque wrenches and shears could always be located in an instant, and that they could never be erroneously used outside of a specific set of operational parameters. That?s the promise of IoT-enabled connected tooling, delivering enormous improvements in operator productivity.

Wearables: Ensuring worker safety at all times:

Wearable technology might be closely associated with the consumer sector throughthe use of fitness monitors, but it is also delivering enormous benefit in industrial environments. In manufacturing, for instance, wearables are increasingly being used to ensure worker safety, with body-worn connected sensors being used to monitor environmental conditions and to provide insight into vital signs such as temperature, pulse and respiration rate.

By embedding personal protective equipment with sensors or radio-frequency identification technology, they become edge devices in the Industrial Internet of Things, collecting and transmitting data to provide insight.

These connected worker platforms aim to more effectively manage worker safety, especially in manufacturing plants which have employees perform tasks alone or handle potentially hazardous substances. It also acts as a means of lowering an organisation?s compliance and administrative costs. Meanwhile, wearables are also being used within manufacturing for ergonomic reasons, to reduce the toll that physical activities take on its workers bodies.

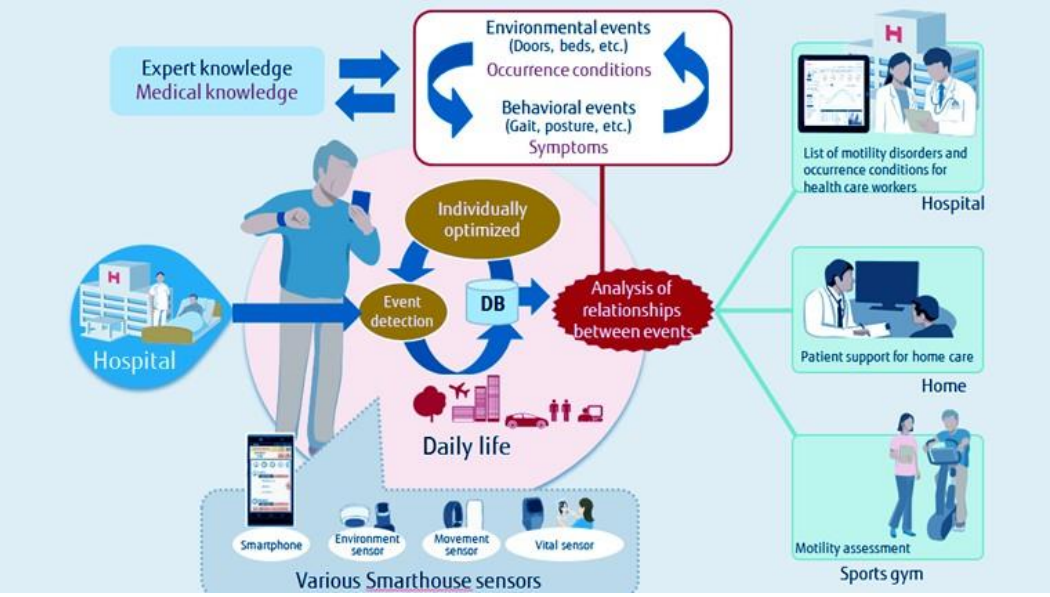
Lecture 8

Health and Lifestyles:

Health:

The healthcare industry is in a state of great despair. Healthcare services are costlier than ever, global population is aging and the number of chronic diseases are on arise. A new paradigm, known as the Internet of Things (IoT), has an extensive applicability in numerous areas, including healthcare. The full application of this paradigm in healthcare area is a mutual hope because it allows medical centers to function more competently and patients to obtain better treatment.

With the use of this technology-based healthcare method, there are unparalleled benefits which could improve the quality and efficiency of treatments and accordingly improve the health of the patients.



Simultaneous reporting and monitoring:

Real-time monitoring via connected devices can save lives in event of a medical emergency like heart failure, diabetes, asthma attacks, etc. With real-time monitoring of the condition in place by means of a smart medical device connected to a smartphoneapp, connected devices can collect medical and other required health data and use the data connection of the smartphone to transfer collected information to a physician. Center of Connected Health Policy conducted a study that indicates that there was a 50% reduction in 30-day readmission rate because of remote patient monitoring on heart failure patients. The IoT device collects and transfers health data: blood pressure, oxygen and blood sugar levels, weight, and ECGs. These data are stored in the cloud and can be shared with an authorized person, who could be a physician, your insurance company, a participating health firm or an external consultant, to allow them to look at the collected data regardless of their place, time, or device.

End-to-end connectivity and affordability:

IoT can automate patient care workflow with the help healthcare mobility solution and other new technologies, and next-gen healthcare facilities. IoT in healthcare enables interoperability, machine-to-machine communication, information exchange, and data movement that makes healthcare service delivery effective. Connectivity protocols: Bluetooth LE, Wi-Fi, Z-wave, ZigBee, and other modern protocols, healthcare personnel can change the way they spot illness and ailments in patientsand can also innovate revolutionary ways of treatment. Consequently, technology-driven setup brings down the cost, by cutting down unnecessary visits, utilizing better quality resources, and improving the allocation and planning.

Data assortment and analysis:

Vast amount of data that a healthcare device sends in a very short time owing to their real-time application is hard to store and manage if the access to cloud is unavailable. Even for healthcare providers to acquire data originating from multipledevices and sources and analyze it manually is a tough bet. IoT devices can collect, report and analyses the data in real-time and cut the need to store the raw data. This all can happen overcloud with the providers only getting access to final reports with graphs. Moreover, healthcare operations allow organizations to get vital healthcareanalytics and data-driven insights which speed up decision-making and is less proneto errors.

Tracking and alerts:

On-time alert is critical in event of life-threatening circumstances. Medical IoT devices gather vital data and transfer that data to doctors for real-time tracking,while dropping notifications to people about critical parts via mobile apps and other linked devices. Reports and alerts give a firm opinion about a patient?s condition, irrespective of place and time. It also helps make well-versed decisions and provide on-time treatment. Thus, IoT enables real-time alerting, tracking, and monitoring, which permits hands-on treatments, better accuracy, apt intervention by doctors and improve complete patient care delivery results.

Remote medical assistance:

In event of an emergency, patients can contact a doctor who is many kilometersaway with a smart mobile apps. With mobility solutions in healthcare, the medics can instantly check the patients and identify the ailments on-the-go. Also, numerous healthcare delivery chains that are forecasting to build machines that can distribute drugs on the basis of patient?s prescription and ailment-related data available via

linked devices. IoT will Improve the patient?s care In hospital. This in turn, will cut on people?s expanse on healthcare

Ingestible sensors:

Ingestible sensors are genuinely a modern-science marvel. These are pill-sized sensors which monitor the medication in our body and warns us if it detects any irregularitiesin our bodies. These sensors can be a boon for a diabetic patient as it would help in curbing symptoms and provide with an early warning for diseases. Proteus Digital Health is one such example.

IoT in LifeStyle:

Wearable IoT devices that allow non-invasive and continuous monitoring of physiological parameters can help in continuous health and fitness monitoring. These wearable devices may can be in various forms such as belts and wrist-bands. The wearable devices form a type of wireless sensor networks called body area networks in which the measurements from a number of wearable devices are continuous sent to amaster node (such as a smart-phone) which then sends the data to a server or a cloud-based back-end for anal; and archiving. Health-care providers can analyze the collected health-care data to determine any health conditions or anomalies.

Commonly uses body sensors include: body temperature, heart rate, pulse oximeter oxygen saturation (SPo2), blood pressure, electrocardiogram (ECG), movement(with accelerometers), and electroencephalogram (EEG). An ubiquitous mobility

approach for body sensor networks in health-care is proposed in. In , a wearable ubiquitous health-care monitoring system is presented that uses integrated electrocardiogram (ECG), accelerometer and oxygen saturation (Spo_2) sensors. Fitbit wristband is a wearable device that tracks steps, distance, and calories burned during the day and sleep quality at night.

Wearable Electronics:

Wearable electronics such as wearable gadgets (smart watches, smart glasses, wristbands, etc.) and fashion electronics (with electronics integrated in clothing and accessories, (e.g.. Google Glass or Moto 360 smart watch) provide various functions and features to assist us in our daily activities and making us lead healthy lifestyles. Smart watches that run mobile operating systems (such as Android) provide enhanced functionality beyond just timekeeping. With smart watches, the users can search the Internet, play audio/video files, make calls (with or without paired mobile phones),play games and use various kinds of mobile applications. Smart glasses allows usersto take photos and record videos, get map directions, check flight status, and searchthe Internet by using voice commands. Smart shoes monitor the walking or runningspeeds and jumps with the help of embedded sensors and be paired with smart- phones to visualize the data . Smart wristbands can track the daily exercise and calories burnt.

Hearables:

Hearables are new-age hearing aids which have completely transformed the way people who suffered hearing loss interact with the world. Nowadays, hearables are compatible with Bluetooth which syncs your smartphone with it. It allows you tofilter, equalize and add layered features to real-world sounds. Doppler Labs is the most suitable example of it.

Moodables:

Moodables are mood enhancing devices which help in improving our mood throughout the day. It may sound like science fiction, but it?s not far from reality.Thync and Halo Neurosciences are already working on it and has made tremendous progress. Moodables are head-mounted wearables that send low-intensity current to the brain

which elevates our mood.

Computer vision technology:

Computer vision technology along with AI has given rise to drone technology which aims to mimic visual perception and hence decision making based on it. Drones like Skydio use computer vision technology to detect obstacles and to navigate around them. This technology can also be used for visually impaired people to navigate efficiently.
